Problem Statement

Background

Agriculture plays a vital role in food production and economic development. Farmers often face challenges in selecting the most suitable crop due to changing weather conditions, varying soil properties, and limited access to scientific recommendations. Incorrect crop selection can lead to reduced productivity, financial losses, and inefficient use of natural resources.

Problem

Traditional farming decisions are mainly based on experience rather than data-driven analysis. As climate conditions continue to change, these methods become less reliable.

The major challenges include:

- Incorrect crop selection
- Low agricultural productivity
- Poor utilization of soil nutrients
- Water resource wastage
- Financial losses for farmers

Proposed Solution

OptiCrop is an AI-powered crop recommendation system that analyzes soil nutrients and environmental conditions using Machine Learning.

The system recommends the most suitable crop based on:

- Nitrogen
- Phosphorus
- Potassium

- Temperature
- Humidity
- pH
- Rainfall

Goals

- Improve crop productivity.
- Increase farmers' profits.
- Promote sustainable agriculture.
- Enable data-driven farming decisions.